KK Thuwajit

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Education

University of California, Los Angeles — Los Angeles, CA
Master of Science in Computer Science — *Sep 2026 – Expected 2027*

University of Wisconsin–Madison — Madison, WI
B.S. Computer Science and Mathematics, GPA 4.00, Hilldale Fellowship, Sophomore Research Fellowship — *Aug 2022 – Dec 2025*

Experience

American Family Insurance — Madison, WI

Machine Learning Engineer Intern — *Jun – Dec 2026*

- **Built the company's first prompt-injection defense, beating Google Model Armor**: 0.8 and 0.6 PR-AUC on unseen public and red-team attacks at 100:1 imbalance, where Model Armor scored under 0.3
- **Comparable 10ms/page latency at 1/10 the cost**, a GCP A100 instance; in contention for deployment
- **Red-teamed a production LLM pipeline** that adjusters use on 100+ page demand packages, from an open brief with no threat model, showing that nearly invisible text makes it report liability factors that do not exist
- **Designed a dataset-agnostic training method** that doesn't assume attacks from the target documents, learning foreign text by splicing other documents into them, making it generalizable to any document

Machine Learning Engineer Intern — *May – Dec 2025*

- **Built a knowledge graph over 100+ page demand letters**: LLM dynamically generates schema per document type and fills them with content into a graph; "list all the patient's symptoms and when each was diagnosed" answers in one query
- **Halved document-splitting errors for said demand letters** with automated prompt optimization

Machine Learning Engineer Intern — *May – Aug 2024*

- **Built LayoutLM document Q&A** and extracted invoice entities from OCR at 95% F1 with LLM post-processing

Selected Projects

ModernLayoutLM: long-context layout-aware document encoder | *PyTorch, Hugging Face* | model — Jul – Oct 2026

- **8,192-token context, 16× LayoutLMv1's, at matching performance**: within 0.03 F1 on FUNSD, CORD and SROIE
- **Pretrained on 11M decontaminated pages** in 28 hours on a single RTX 3090

llama-router, a per-GPU model scheduler for llama.cpp | *Python, llama.cpp, Docker, Nix* | code — Jul 2026 – Present

- **Served 99.6% of 10,000 requests at 256 concurrent** across three swapping models with none dropped except for client timeouts, where llama.cpp and Ollama freeze or drop requests when users switch models mid-request
- **Implemented a queue for LLM request serving**: uses an LRU queue to minimize LLM swap time by passing through an OpenAI-compatible endpoint; in production on two GPU servers

GraphIV, an open map of arXiv for research agents | *PostgreSQL, PyTorch, MCP* | demo, code, data — May 2026 – Present

- **Released a biweekly-updating citation-intent graph of arXiv**: 3.17M papers, 74.8M citations intents labeled (extends, critiques, compares and more), at 1,762 rows/s by a DistilBERT distilled from an LLM teacher
- **Designed BoundedField, an LLM-free citation-graph retriever at 0.81 recall@1000** against 0.37 for embedding search, measured on real reference lists of human-written literature reviews on arXiv
- **Designed LeyLine, a Monte Carlo miner of a field's lineages over intent-labeled citations**, finding 136 of 557 papers too new to be cited that later gained popularty, where a pure search-and-rerank system reached none
- **Serves LLM agents over a 10-tool MCP server** writing grounded literature reviews, with a public dashboard/demo

Homelab, infrastructure as code for three machines | *NixOS, ZFS, disko, sops, Tailscale, Caddy* | code — Oct 2025 – Present

- **Architected geo-redundant infrastructure across Los Angeles and Bangkok**, with DNS resolvers at both sites and two-tier backups: 6-hourly offsite sync to a RAIDZ1 ZFS array, snapshots retained up to 3 months
- **Managed CI/CD for a Siriraj Hospital cancer research lab's website**: updating the lab's publications and labels via self-hosted Forgejo runners from PubMed, classifying each with a self-hosted LLM, and publishing them weekly
- **Deployed GraphIV and llama-router to production as NixOS modules** alongside 26 self-hosted services, all declared in one flake whose templates generate each service's reverse proxy and storage provisioning

Research

Waisman Center, UW–Madison, brain imaging research assistant, May 2023 – 2026. Cut MRI scan time from 9 to 2 minutes with jointly trained U-Net regularizers; first author, *ISMRM* 2024

Madison Experimental Mathematics Lab, research fellow, Feb – Dec 2025. Reduced ensemble Kalman filter cost from cubic to quadratic with PPO agents that generalize to unseen systems

VISTEC, Thailand, biosignal research assistant, May 2020 – May 2023. Multiscale CNNs for EEG seizure detection at 95% accuracy and PPG respiratory sensing: EEGWaveNet, *IEEE Trans. Industrial Informatics* 2021, first author; RRWaveNet, *IEEE Internet of Things Journal* 2023, co-corresponding author; ApSense, *IEEE Internet of Things Journal* 2024, co-author

Skills

ML: PyTorch, Hugging Face, llama.cpp, distillation, retrieval and evaluation design, LLM agents and MCP
Systems: Python, SQL, PostgreSQL/pgvector, Neo4j, Docker, Nix/NixOS, Linux, ZFS, JavaScript